Analysis of Forest-Based Tornadoes Using Treefall Patterns

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ABSTRACT

Many tornadoes occur in rural forested regions of the world where traditional damage indicators are often unavailable to determine, or may underestimate, a tornado's true intensity. Future versions of the EF scale are expected to include guidelines for treefall pattern matching to determine tornado intensity. This study explores the reliability and uncertainty of using tornado treefall patterns as a method for assessing the maximum 3-second gust speed and swirl ratio of tornadoes in forested regions. Through mathematical analysis, treefall patterns are determined to depend primarily on the ratio of the tornado's radial, tangential and translational speeds, as well as the critical tree failure speed. Moreover, four typical types of tornado treefall patterns are differentiated based on vortex-tree failure locations relative to the tornado's radius of maximum wind speed. As a result, when estimates for the critical tree failure and tornado translation speeds are known, treefall patterns can be consistently linked to an estimate of a tornado's maximum wind speed and swirl ratio. The damage paths of four tornadoes, including the previously analyzed Alonsa, MB EF4 tornado, are analyzed using custom software implementing a Monte Carlo simulation for treefall pattern matching to estimate the maximum 3-second gust speed and swirl ratio, as well as the associated uncertainty. Of the three newly analyzed events, two are estimated to have higher EF Scale ratings than those suggested by the current Canadian EF-scale.

SIGNIFICANCE STATEMENT

Current methods in the Canadian EF scale are often unable to rate tornadoes above EF2 when considering only tree damage. However, previous studies have shown that treefall pattern matching has the potential to provide higher ratings. Building on previous works, this study explores the reliability and uncertainty in using treefall pattern matching for estimating a tornado's maximum 3-second gust speed and swirl ratio. Through the use of a Monte Carlo simulation approach, improved estimates of the range of tornado windspeeds are obtained.

1. Introduction

Canada is home to 270 million hectares of boreal forest – over 60% of Canada's land cover (Brandt 2009) – much of which is frequently hit by tornadoes. The Northern Tornadoes Project (NTP), a program run by the Canadian Severe Storms Laboratory, has the goal of assessing and documenting every tornado that occurs in Canada. Figure 1 shows the locations of tornadoes from 2017 to 2024 that have occurred in the Canadian boreal zone (NTP 2020). This analysis shows that 26% of tornadoes recorded by the NTP have occurred in the Canadian boreal zone. Butt et al. (2024) also found that these northern forested areas tend to be sparsely populated with the only observable indication of damage often being snapped or uprooted trees. Overall, considering the frequency of tornadoes in forested regions of Canada, properly assessing tree damage is critical for accurately determining Canada's tornado climatology. Moreover, this is also important for remote forested areas of the U.S. and northern Europe (Karstens et al. 2013; Shikhov and Chernokulsky 2018).

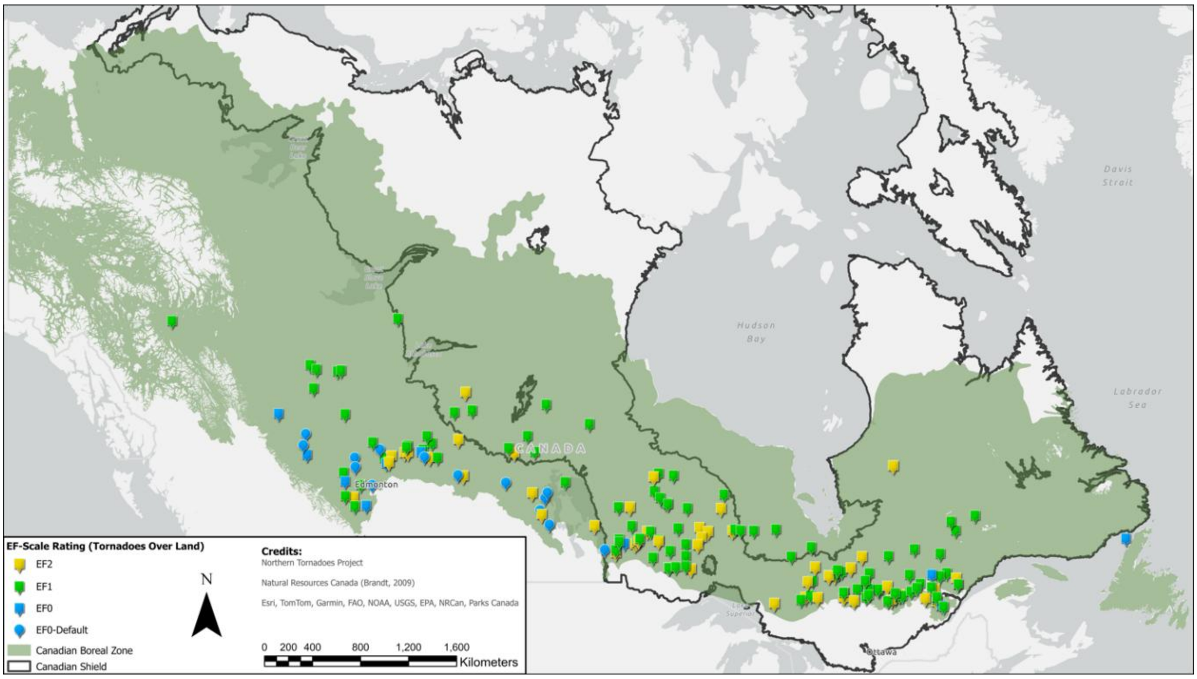

Figure 1: A map of Canadian tornadoes (over land) from 2017-2024 that occurred in the Canadian boreal zone, along with a reference to the location of the Canadian Shield.

The intensity of tornadoes in Canada is assessed using the Enhanced Fujita (EF) scale, which rates tornadoes on a scale from EF0 to EF5 based on their estimated maximum wind speed (McDonald and Mehta 2006; Sills et al. 2014). The Canadian EF Scale (NTP 2024), which is a modified version of the U.S. EF Scale, estimates wind speeds based on 31 Damage Indicators (DIs), mostly representing different structures, as well as trees. Each DI includes Degrees of Damage (DOD) that outline the sequential damage stages observed. Every DOD is associated with upper-bound (UB), lower-bound (LB), and expected (EXP) wind speeds. In damage surveys, the worst failure mode is identified and linked to a DOD category, and a determination is made on whether the upper-bound, lower-bound, or expected value is more appropriate based on guidance in the document related to construction style, quality, and age. These factors help determine the estimated wind speed responsible for the damage.

Currently, the Canadian EF scale (NTP 2024) has a DI for trees that differs from the tree DIs in the U.S. EF scale. Whereas the U.S. EF scale provides wind speed estimates for individual (single) hardwood and softwood trees, in the Canadian EF Scale DODs are based on the percentage of mature trees snapped or uprooted in a given area of the tornado's damage track. These percentages are estimated using a method referred to as the "Scalable Box Method" that was developed by researchers at the NTP (Sills et al. 2020). Godfrey and Peterson (2017), among others, have also aimed to relate the percentage of trees downed in fixed-sized boxes as a function of wind speed. Table 1 shows the Degrees of Damage for trees, along with the guidance provided for this damage indicator.

Of note in Table 1 are restrictions that potentially prevent the accurate assessment of EF3+ tornadoes. In areas of shallow soil, the default is to use the lower-bound wind speed. The Canadian Shield is a geological area of Canada with notably shallow soil (Government of Canada 2019). As seen in Figure 1, many of these forested tornadoes also occur in the Canadian Shield, leading to many of these tornadoes being rated at the lower-bound wind speed. Another problem related to the accurate assessment of EF3+ tornadoes is that the wind speed at which a given tree will fail, referred to as the tree's critical failure wind speed (Karstens et al. 2013; Quine and Gardiner 2007), is difficult to determine, and is generally low relative to the wind speeds of EF3+ tornadoes.

With these restrictions, current methods for assessing tree damage in tornadoes often have the limitation of being unable to estimate wind speeds above EF2, which could lead to the potential underrating of many forest-based tornadoes in Canada and other regions of the world. Assessing the true maximum wind speeds in tornadoes is of critical importance to better design safety measures for communities, create better weather forecasting models, develop a broader understanding of tornado behavior, as well as update building codes and engineering standards. Recently published studies on wind speeds in tornadoes have also noted bias in tornado wind speed estimates (Lombardo et al. 2023; Dahl et al. 2017; Miller et al. 2024). Overall, this leads to the need for alternative methods of assessing tornado treefall damage (Edwards et al. 2013), particularly in forested regions with few, if any, other potential damage indicators.

Table 1: DOD descriptions and wind speed estimates for the Canadian Tree DI, along with the DI notes, as adopted by the Northern Tornadoes Project (2024).

DOD	Description of Damage	Wind Speed Estimates, ms^{-1} & (km/h)		
		Lower-Bound (LB)	Expected (EXP)	Upper-Bound (UB)
1	Small limbs broken (up to 5 cm diameter)	15.3 (55)	19.4 (70)	23.6 (85)
2	Large limbs broken (greater than 5 cm diameter)	18.1 (65)	25 (90)	30.6 (110)
3	Up to 20% of mature trees snapped and/or uprooted	22.2 (80)	31.9 (115)	41.7 (150)
4	More than 20% of mature trees snapped and/or uprooted	29.2 (105)	41.7 (150)	52.8 (190)
5	More than 50% of mature trees snapped and/or uprooted	40.3 (145)	52.8 (190)	63.9 (230)
6	More than 80% of mature trees snapped and/or uprooted; numerous trees may be denuded/debarked by missiles with only stubs of largest branches remaining.	52.8 (190)	65.3 (235)	76.4 (275)

Notes:

- To estimate the percentage of trees snapped and/or uprooted, use the NTP Scalable Box Method.

- For satellite-based assessments of forest damage (having limited imagery resolution, lack of tree health/species information, lack of plantation/age information), the NTP defaults to the lower-bound wind speed – with the following exceptions:
    - DOD6, deep soil and close to 100% treefall (EXP-20 km/h)
    - DOD5, deep soil and close to 80% treefall (EXP-20 km/h)
    - No leaves on trees in spring or fall (EXP)

- For all other assessment types, the NTP defaults to the expected wind speed – with the following exceptions:
    - Very shallow soil (LB)
    - Poor tree health (disease/infestation, broken at base of trunk due to girdling, etc.), shallow root balls, even-aged monoculture plantation (LB)
    - Most trees snapped in area without deep soil (EXP-20 km/h)
    - Forests or woodlots composed of mature, deep-rooted red oak, red maple, beech, hemlock or white cedar (EXP+20 km/h)
    - No leaves on trees in spring or fall and very shallow soil (EXP)
    - No leaves on trees in spring or fall and soil between very shallow and deep (EXP+20 km/h)
    - No leaves on trees in spring or fall and deep soil (UB)

The first work to examine this was Letzmann (1923), who noted that forest-based tornadoes tend to leave specific treefall patterns along the tornado's path and developed a simple analytical method using the Rankine vortex model for reconstructing similar patterns. Since then, significantly more research has been conducted into simulating tornado wind fields and reconstructing treefall patterns as a method to estimate the maximum wind speed of tornadoes. Table 2 provides a summary of the papers that have advanced the approaches for identifying and using tornado damage patterns to trees and forests to estimate tornado wind speeds.

Table 2. Summary of previous research using damage patterns to trees and forests.

Authors	**Events Analyzed**	**Description**
Letzmann (1923); Review and English Translation by Peterson (1992)	N/A	Pioneering work linking treefall damage swaths to tornado wind fields. Utilizes Rankine vortex model; demonstrates how treefall patterns change for different vortex parameters.
Holland et al. (2006)	N/A	Presents framework combining Letzmann's model with a simplified treefall model to produce systematic treefall patterns for different vortex parameters and critical tree failure speeds.
Beck and Dotzek (2010)	Milosovice, Czech Republic (2001) – F3 Castellcir, Spain (2006) – F2	Uses a method similar to Holland et al. to manually match treefall swaths for estimating tornado wind fields. Estimates uncertainty using a range of critical tree failure speeds.
Karstens et al. (2013)	Joplin, Missouri, US (2011) – EF5 Tuscaloosa–Birmingham, Alabama, US (2011) – EF4	Used aerial imagery to acquire individual treefall directions. Presents a more efficient simulation method, similar to Holland et al., to manually match treefall damage swaths and patterns.

Lombardo et al. (2015)	Joplin, Missouri, US (2011) – EF5	Compared treefall-based and structural-based methods for wind field estimation of the Joplin tornado.
Rhee and Lombardo (2018)	Naplate, IL, US (2017) – EF3 Tuscaloosa–Birmingham, Alabama, US (2011) – EF4 Sidney, Illinois, US (2016) – EF2	Utilized a grid-search approach to automatically find the best-matching treefall patterns. Analyzed both tree and crop damage.
Chen and Lombardo (2019)	Joplin, Missouri, US (2011) – EF5	Presents an analytical method using numerical optimization to more efficiently find the best-matching patterns.
Stevenson et al. (2023)	Alonsa, Manitoba, Canada (2018) – EF4	Compared the method of Rhee and Lombardo (2018) to structural damage.

As a high-level overview, current treefall pattern-matching methods work by simulating a tornado vortex travelling over the same location and in the same direction as the observed tornado (Karstens et al. 2013; Rhee and Lombardo 2018). A simulated treefall pattern is calculated using analytical models of the tornado wind field and tree failure mechanics, with treefall occurring at the first instance that the critical wind speed is exceeded for every tree. From there, the observed treefall pattern is compared to the simulated pattern, with the similarity between these patterns taken as proportional to the similarity between the characteristics of the observed and simulated tornado-vortex wind field. After simulations of many patterns, the best matching patterns can be utilized to estimate the wind speeds and characteristics of the observed tornado.

Holland et al. (2006), Beck and Dotzek (2010), and Karstens et al. (2013), building on the work of Letzmann (1923), demonstrated the use of analytical vortex models for simulating treefall patterns, with Karstens et al. (2013) and Beck and Dotzek (2010) applying them to analyze various tornadoes throughout the US and EU respectively. These works laid the foundation for utilizing treefall pattern matching as a method for estimating tornado wind fields and maximum wind speeds. Furthermore, these works generally found consistent EF-scale ratings between structural and treefall-based wind speed estimates. For validation of treefall pattern-based methods, comparison between structural and treefall estimates for the Joplin, Missouri tornado of 2011 was performed by Kuligowski et al. (2014) and Lombardo et al.

(2015). Similarly, Stevenson et al. (2023) applied the treefall pattern matching methodology from Rhee and Lombardo (2018) to the Alonsa, Manitoba EF4 tornado, which produced similar estimates to that of the observed structural damage upon which the EF4 rating was based.

There have been many recent improvements to algorithms for implementing treefall pattern-matching methods. Rhee and Lombardo (2018) developed a grid search approach giving more accurate results for estimating vortex parameters based on the best matching patterns. Chen and Lombardo (2019) proposed an analytical method for solving treefall patterns (in contrast to simulation) for two distinct tornado wind field models – the Rankine and Burgers-Rott vortices (Burgers 1948; Rott 1958) – and utilized a numerical optimization approach to fit best-matching treefall patterns. This method is significantly faster than the simulations of Rhee and Lombardo (2018). Additionally, it can consider multiple transects when performing numerical optimization in a piece-wise approach, to better fit the overall extent of the treefall damage. Generally, these methodologies are limited in their ability to assess the uncertainty and variability in their results.

In general, state-of-the-art techniques for wind speed estimation now tend to consider the variability of parameters by creating fragility curves, often using Monte Carlo simulations. Examples include Ellingwood et al. (2004), Amini and van de Lindt (2014), Roueche et al. (2016), and Gavanski and Kopp (2017) for wood-frame houses. Godfrey and Petersen (2017) used an approach which considered the fraction of trees down within a particular zone (but not combined with a tornado wind field model). Miller et al. (2024) used tornado wind field models to assess the flight of large debris by combining wind field model parameters in a Monte Carlo simulation to estimate the range of tornado vortex parameters that could lead to the observed flight distances. Analysis of treefall patterns, as described herein, falls into this same type of approach; however, there have been no studies to date that have utilized the Monte Carlo simulation – fragility curve approach. One advantage of this type of framework is that it allows new knowledge to be incorporated as it becomes available in straight-forward ways, allowing the variation to be modeled and accounted for.

The objective of this study is to analyze tornado treefall patterns and assess their uncertainty in determining maximum wind speeds, through the use of Monte Carlo simulations that consider the variation due to different wind field models, critical wind speeds for treefall, vortex translational speeds, and location and scale of treefall pattern transects. Additionally, novel analysis of analytical treefall pattern solutions provides a greater understanding of how the patterns are related to their tornado wind fields, creating insight into damage observations. Sections 2 - 3 discuss and examine the analytical vortex models and the relationships between vortex parameters and types of treefall patterns, building off the work of the previously mentioned studies. Furthermore, using a Monte Carlo simulation, a novel and more computationally efficient matching methodology is constructed. This methodology, discussed in section 4, is better suited to assessing uncertainty. Then in sections 5, the resulting method is applied to four Canadian forest-based tornadoes to examine the outcomes of the approach.

2. Tornado Wind Field Models

Tornado wind field models can range from simple analytical vortex models like the Rankine vortex, to more intricate models such as simplified solutions of the Navier-Stokes equations (Burgers, 1948; Rott, 1958; Baker and Sterling, 2017) and the semi-empirical Twisdale model (Twisdale et al. 2023; Dunn and Twisdale, 1979). These are used in a range of applications beyond assessing tornado damage to trees and forests. For example, the Twisdale model is the basis for the tornado wind speed maps in ASCE 7-22 (2022). The Baker-Sterling model has been used to fit the data for tornado-vortex simulator wind loading results (Brusco and Kopp, 2024). These models represent the state-of-the-art for such varied analyses, although there is recognition that they are significant simplifications and more work is needed (Haan et al., 2024).

a. Vortex Field Definition

For the purposes of treefall patterns, a tornado vortex is simulated as a two-dimensional projection of the overall vortex field near ground level due to the lower height of trees relative to the overall height of a tornado vortex (Karstens et al. 2013). The two-dimensional projection of a tornado vortex field is defined at any point as the sum of the radial ($\vec{V}_{rad}$), tangential ($\vec{V}_{tan}$), and translational ($\vec{V}_s$) velocities (Chen and Lombardo, 2019). For the purpose of computing a treefall pattern, the vortex is assumed to translate in a constant direction and with constant parameters for a short duration over a chosen transect of the tornado's damage path. For simplicity, translation in the positive y-direction was chosen, in common with Karstens et al. (2013), but by rotating, any other direction can also be used. The strengths of the radial and tangential components are calculated by multiplying the unitless radial and tangential velocity profile functions ($P_r,\ P_t$) by maximum radial and tangential velocity constants ($V_r,\ V_t$). Due to the biomechanics of tree failures being mostly bending-moment-based overturning (uprooting) or snapping, the vertical velocity component of the vortex is assumed to be a negligible contribution to the failure of a tree in this model. Generally, single-core vortex velocity profile functions are assumed to be symmetric about any rotation and depend only on the distance from the center of the vortex (Kim and Matsui, 2017).

$$\frac{V}{V_{max}} = P\left(\frac{r}{R_{max}}\right) \tag{1}$$

The unitless version of these profiles, as shown in equation (1), specifies the ratio of velocity to maximum velocity, (V/V_{max}), as a function of radial distance from the origin to the radius at which the maximum velocity occurs, (r/R_{max}). Figure 2 shows an example of a generated two-dimensional vortex field along with a diagram of the velocity components.

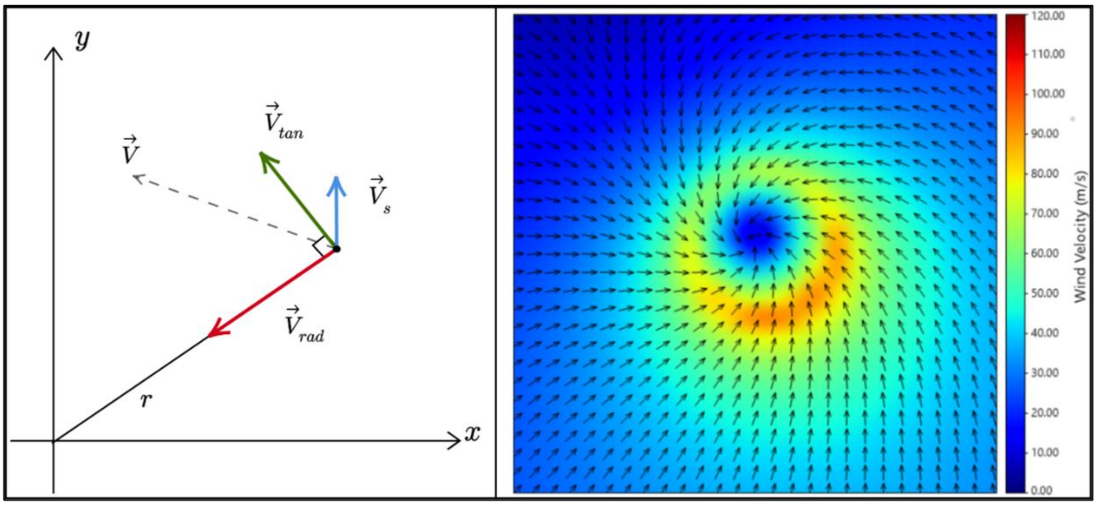

Figure 2: A diagram depicting the components of the two-dimensional vortex wind velocity (left) and a generated two-dimensional vortex field (right) showing an example vortex field.

From here, the radial, tangential and translational unit vectors can be calculated as:

$$r = \sqrt{x^2 + y^2} \tag{2}$$

$$\hat{V}_{rad} = \left[\frac{-x}{r}, \frac{-y}{r}\right],\ \hat{V}_{tan} = \left[\frac{-y}{r}, \frac{x}{r}\right],\ \hat{V}_s = [0, 1] \tag{3}$$

Using the sum of the vectors, the overall wind velocity vector $(\vec{V})$ at any given point, (x, y), can be calculated as:

$$\vec{V} = \vec{V}(x, y) = \vec{V}_{rad} + \vec{V}_{tan} + \vec{V}_s \tag{4}$$

$$\vec{V} = P_r V_r \hat{V}_{rad} + P_t V_t \hat{V}_{tan} + V_s \hat{j} \tag{5}$$

$$\vec{V} = \left[-\frac{P_r V_r x + P_t V_t y}{r}, \frac{P_t V_t x - P_r V_r y}{r} + V_s \right] \tag{6}$$

b. Single-Core Vortex Profiles

For this model, the vortex profiles utilized are single-core, axisymmetric, unitless, and normalized such that their maximum of 1 occurs when $r = R_{max}$. As shown in Figure 3, normalized profiles start at zero and are increasing on the interval $0 < rR_{max}^{-1} < 1$, and decreasing on the interval $rR_{max}^{-1} > 1$. A commonly used profile is the Modified Rankine vortex profile, equation (7), originally proposed for tornadoes by Letzmann (1923) and utilized in numerous other studies including Karstens et al. (2013), Rhee and Lombardo (2018), and Chen and Lombardo (2019).

$$P_t = P_r = \begin{cases} (rR_{max}^{-1})^{\varphi}, & if\ r \leq R_{max} \\ (r^{-1}R_{max})^{\varphi}, & if\ r > R_{max} \end{cases}, \qquad 0.5 \leq \varphi \leq 1 \tag{7}$$

As noted above, other vortex profiles exist. For example, a more recently proposed vortex profile is the Baker-Sterling model (Baker and Sterling 2017), which simplifies to the Bjerknes vortex profile, equation (8), when normalized and converted to be two-dimensional (Bjerknes 1921).

$$P_t = P_r = \frac{2rR_{max}}{r^2 + R_{max}^2} \tag{8}$$

The Burgers-Rott profile, equation (9), is based on the Burgers-Rott vortex (Burgers 1948; Rott 1958), which is an exact solution of the Navier-Stokes equations. Unlike the Rankine and Bjerknes models, the Burgers-Rott radial and tangential profiles differ. The Burgers-Rott radial profile is assumed to scale linearly indefinitely, which is not indicative of a tornado and, as such, can be replaced for tornado modelling with either its tangential profile or the linear Rankine profile where $\varphi = 1$.

$$P_t = \frac{k_1[1 - \exp(-k_2 r^2 R_{max}^{-2})]}{2\pi r R_{max}^{-1}}, \qquad k_1 = 8.7836, \qquad \mathrm{k}_2 = 1.2564 \tag{9}$$

An extension of the Burgers-Rott profile is the Sullivan profile, equations (10/11) (Sullivan 1959). Like Burgers-Rott, the radial profile is assumed to scale linearly indefinitely and can be similarly replaced. Figure 3 gives an overview of the graphs for each mentioned profile, all of which are utilized in the methodology of this study.

$$P_t = \frac{k_1}{rR_{max}^{-1}} H(k_2 r^2 R_{max}^{-2}), \qquad k_1 = 0.029890, \qquad k_2 = 6.2381 \tag{10}$$

$$H(x) = \int_0^x \exp\left(-t + 3\int_0^t \frac{1-\exp(-s)}{s} ds\right) dt \tag{11}$$

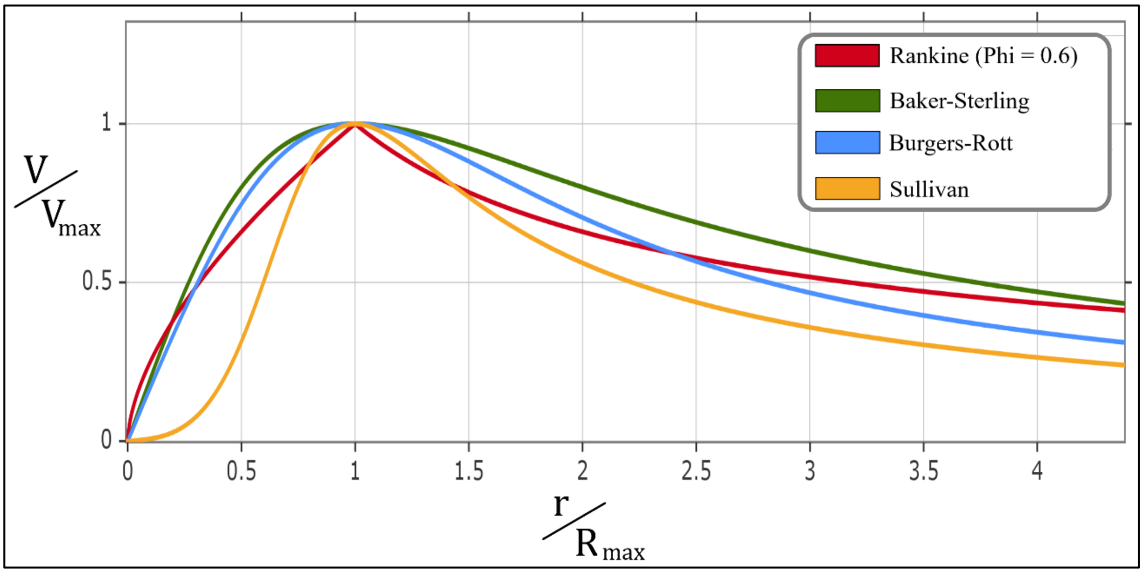

Figure 3: Normalized, unitless, Modified Rankine, Baker-Sterling (Bjerknes), Burgers-Rott, and Sullivan profiles.

c. Tornado Translational Speed Estimation

The tornado translation speed (V_s) can be estimated from observed storm characteristics, typically from available radar data (Beck and Dotzek 2010; Karstens et al. 2013; Rhee and Lombardo 2018) but is difficult to determine without time-stamped visual evidence. However, it can be reasonably assumed that a tornado will move at the same speed as its parent mesocyclone. If the tornado occurred within the range of a weather radar station where Doppler radial velocity data is available, then such velocity data can be used to track the movement of the center of the mesocyclone over the time intervals that the tornado was present. In cases where radial velocity data are not available, the presence and location of the mesocyclone can be inferred from reflectivity features, such as the 'hook echo' caused by the mesocyclone.

In the current work, the center of the mesocyclone is estimated at multiple points in separate radar scans which are then used to calculate the distance that the mesocyclone travelled. By assuming the tornado travelled along a quasi-linear path, the total distance travelled is then divided by the total time between radar scans to estimate the average translation speed.

d. Maximum Velocity and 3-Second Gust Speed

As previously described, the maximum wind velocity (V_{max}), occurs when $r = R_{max}$. When summing a pair of vectors, the magnitude of their sum will be maximized if both vectors are in the same direction. Using these facts, finding the point of maximum velocity is the same as finding the point on the circle, $x^2 + y^2 = R_{max}^2$, where $\vec{V}_{rad} + \vec{V}_{tan}$ is in the same direction as $\vec{V}_s$. Since, for this model, $\vec{V}_s$ is in the positive y-direction, it is implied that the x-component of the $\vec{V}_{rad} + \vec{V}_{tan}$ must equal 0 and the y-component must be positive. From there it follows that V_{max} occurs at the point:

$$x^* = \frac{V_t R_{max}}{\sqrt{V_t^2 + V_r^2}}, \qquad y^* = -\frac{V_r R_{max}}{\sqrt{V_t^2 + V_r^2}} \tag{12}$$

Furthermore, V_{max} can be shown to be a function of V_r, V_t, V_s, as:

$$V_{max} = V_s + \sqrt{V_t^2 + V_r^2} \tag{13}$$

By examining the swirl ratio, $S = V_t V_r^{-1}$, using the Baker and Sterling (2017) definition, and equation (12), the location of V_{max} can be expressed as an angle offset (α) from the y-axis, along the R_{max} circle as shown in Figure 4. Notably, this angle offset (α) is also the angle between the radial and the sum of the radial and tangential velocity components as discussed in Letzmann (1923), Holland et al. (2006) and Rhee and Lombardo (2018). Generally, this implies that the maximum wind velocity does not occur along the central axis of the vortex. In the literature, there are different definitions for the swirl ratio; however, this definition was used due to the two-dimensional vortices considered in this study and based on the discussion in Haan et al. (2024); see also (Twisdale et al. 2023; Dunn and Twisdale, 1979).

$$\alpha = \tan^{-1}(S) \tag{14}$$

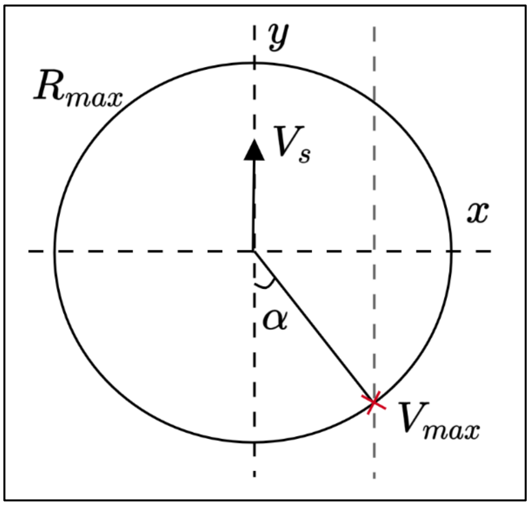

Figure 4: Diagram of the location of maximum wind velocity (V_{max}), as an angle offset from the y-axis, along the R_{max} circle, where α is calculated using equation (14).

Generally, the wind speed estimates for damage given by the EF scale are based on a 3-second-average gust speed, as opposed to an instantaneous wind speed (McDonald, and Mehta 2006). As such, the maximum 3-second gust speed can be approximated by integrating over a 3-second window centered around the location of maximum velocity, and then dividing by the window's area, similar to that used by Miller et al. (2024):

$$V_{3-max} = \frac{\int_{y^*-1.5V_s}^{y^*+1.5V_s} \int_{x^*-1.5V_s}^{x^*+1.5V_s} \left\| \vec{V}(x,y) \right\| dxdy}{(3V_s)^2} \tag{15}$$

3. Computing Analytical Treefall Patterns

a. Critical Tree Failure Speed Estimation

One of the important parameters in the analysis of treefall patterns is the critical tree failure wind speed (V_c). Similar to Chen and Lombardo (2019), as a simulated vortex translates over a simulated forest of trees, the trees are assumed to fail (snap or uproot) on average at a specified critical-tree failure wind speed. Mechanistic models used to calculate the critical tree failure wind speed are generally based on assessing the balance between the destabilizing wind forces on the above-ground elements of trees and the combined resistive forces of the soil, roots and stem. The destabilizing forces are dependent on the wind environment, the location of the tree within the forest canopy, the crown characteristics (e.g. mass, size, and any streamlining [which is the reconfiguration/bending of the branches and leaves to reduce wind resistance, Haller (1990)]) and the stem (trunk) characteristics (e.g. shape, height, and mass). The resistive forces depend on the stem characteristics (e.g. diameter and wood strength) and the root-plate morphology, soil type and state, and the water-table location (Quine and Gardiner 2007; Mansour et al. 2024). Moreover, the resistances of trees to overturning (also known as uprooting or windthrow, Sagi et al., 2019) and stem breakage failure (also known as snapping) are commonly based on empirical relationships found from field static tree pulling and laboratory timber strength tests (e.g. Nicoll et al. 2006).

The current method used to calculate V_c assumes tree overturning, since most of the trees in the field study (described below) were observed to have failed through this mode. The resistance to overturning is based on correlations between the bending moment required to overturn trees and the stem weight or root–soil plate weight (Levy et al. 2004; Nicoll et al. 2006). Thus, the critical wind speed (V_{c-over}) at the forest canopy top was found using the relationship of Gardiner et al. (2000):

$$V_{c-over} = \frac{1}{k.D}\left[\frac{C_{reg}.SW}{\rho.G.d}\right]^{1/2}\left[\frac{1}{f_{edge}.f_{CW}}\right]^{1/2} ln\left(\frac{h-d}{z_o}\right) \tag{16}$$

where k is Von Karman's constant (0.4), d (m) is the zero-plane displacement, z_o (m) is the aerodynamic roughness, D (m) is the average spacing between trees, G is an empirically derived gust factor based on bending moment, h (m) is the tree height, C_{reg} is a regression constant dependent on the tree species, soil and rooting depth, SW (kg) is the stem weight of the tree and ρ (1.226 kg/m^3) is the air density. Factors f_{edge} and f_{CW} account for the tree position relative to the forest edge and additional loads due to the overhanging weight of the crown under bending.

Another important parameter in calculating the critical tree failure speed is the empirical gust factor, G. This captures in a single term the complex influence of the turbulent wind structure in a forest canopy and the dynamic response of trees to fluctuating winds. Although G has been estimated using field and wind tunnel studies for synoptic wind fields (Gardiner et al. 1997), equivalent studies for rotating tornadic winds do not exist. A similar approach to Savory et al. (2001) was adopted, who modelled tornado loading of lattice transmission towers, and a unity value of G was selected, thus assuming a quasi-static response by the trees. This seems reasonable given the expected duration, size (relative to the trees) and frequency of tornadic gusts, and the natural frequency, porosity and very high damping ratio of trees, which is in broad agreement with ASCE 7-22 (American Society for Civil Engineers 2022). Finally, the values of the factors f_{edge} and f_{CW} were also set to unity, assuming the majority of damaged trees were distant from the forest edge and experienced little bending during failure.

b. Analytically Solving Vortex-Tree Failure Locations

In this section, we develop our analytical approach. In this approach, when trees fail, the propagation effects and interactions between trees are ignored, with the underlying assumption that these effects will average out to trees falling in the direction of the wind at the instance of failure. Analytically this assumes that simulated trees will fail at the first encountered position in the vortex field where the magnitude of the wind velocity is greater than the critical-tree failure velocity, referred to in this study as the "vortex-tree failure location". (In the polar equations of Chen and Lombardo, 2019, this is the "critical wind speed curvilinear trace".) This can be solved as:

$$\|\vec{V}(x,y)\| = V_c \tag{17}$$

Substituting in equation (6) for $\vec{V}$ yields:

$$0 = \left(\frac{P_r V_r x + P_t V_t y}{r}\right)^2 + \left(\frac{P_t V_t x - P_r V_r y}{r} + V_s\right)^2 - V_c^2 \tag{18}$$

Often for vortex profiles, equation (18) can only be expressed implicitly as there exist multiple values of y for each value of x where the magnitude of the wind velocity is equal to the critical-tree failure speed. However, trees are assumed to fail at the first encountered location where $\|\vec{V}\| = V_c$. Since the vortex is travelling in the y-direction, trees will fail at the greatest y-value that satisfies equation (18) for a given x-value. As a result, the vortex-tree failure locations are modelled as a function of x, and solved explicitly for the greatest y-value that satisfies equation (18). Mathematically,

$$y = f(x) = \max(\mathcal{R}) \tag{19}$$

where $\mathcal{R}$ is the set of real roots for a given x-value and the set of (constant) parameters, V_r, V_t, V_s, V_c, and R_{max}, as well as selected profiles P_r, and P_t. For example, with the Baker-Sterling (Bjerknes) model, the vortex-tree failure location equation can be expressed as finding the greatest real root of a quartic equation in terms of y in equations (20), (21). In this case, polynomial equations are desired, since their roots can be quickly and reliably solved using numerically algorithms such as the one presented by Yuksel (2022) or using functions in common software packages such as the “roots” function in MATLAB (2024) or the Python NumPy library (2020). An example of a vortex-tree failure location function is depicted in Figure 5.

$$0 = \left(\frac{2R_{max}(V_r x + V_t y)}{r^2 + R_{max}^2}\right)^2 + \left(\frac{2R_{max}(V_t x - V_r y)}{r^2 + R_{max}^2} + V_s\right)^2 - V_c^2 \tag{20}$$

$$\Rightarrow \mathcal{R} = \text{roots}(a_4 y^4 + a_3 y^3 + a_2 y^2 + a_1 y + a_0) \tag{21}$$

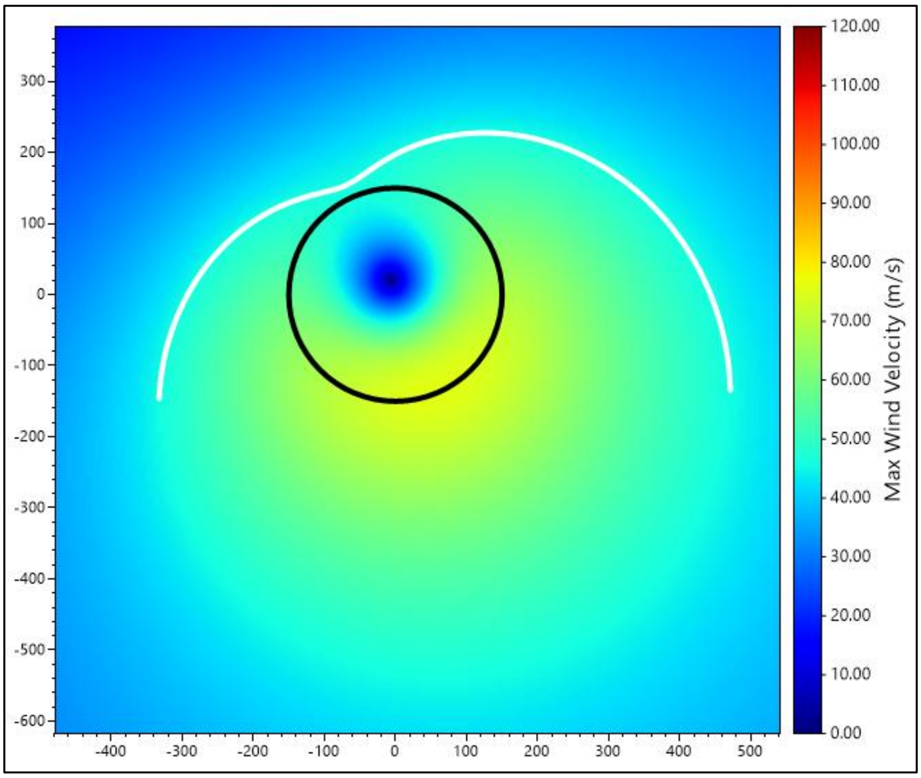

Figure 5: A graph of the vortex-tree failure locations (white curve) using the Bjerknes model with, $V_r = 55ms^{-1}, V_t = 25ms^{-1}, V_s = 15ms^{-1}, V_c = 45ms^{-1}, R_{max} = 150m$ with the $r = R_{max}$ circle in black.

Unfortunately, with other profiles like the Modified Rankine (for arbitrary φ), Burgers-Rott, and Sullivan profiles, the vortex-tree failure location, defined by equation (19), cannot always be expressed as a polynomial. By investigation of equation (18) and the Bjerknes profile, it can be concluded that any profile that is a rational polynomial in rR_{max}^{-1}, where the numerator and denominator polynomials contain only odd and even powered terms respectively, will yield a polynomial for equation (19). As such, the Burgers-Rott and Sullivan tangential profiles are approximated using numerically optimized function fitting as:

$$P_t = \frac{k_1 r R_{max}^{-1} + k_3 r^3 R_{max}^{-3}}{k_0 + k_2 r^2 R_{max}^{-2} + k_4 r^4 R_{max}^{-4}} \tag{22}$$

where, $k_0 = 1.8242, k_1 = 3.1115, k_2 = 1.5189, k_3 = 1.0, k_4 = 0.76843$, and $k_0 = 1.1180$, $k_1 = 0.060702$, $k_2 = -1.0$, $k_3 = 2.2901$, $k_4 = 2.2327$, for the Burgers-Rott and Sullivan profile approximations respectively. These approximate profiles have a worst-case error of approximately 0.01 and 0.02 when compared to the Burgers-Rott and Sullivan profiles, respectively. Moreover, they produce degree 8 and 10 polynomials for equation (19), when the radial profile of the Burgers-Rott and Sullivan models are replaced by their tangential profile and the Rankine profile, respectively.

In the case of the Modified Rankine profile, when $\varphi = 1$, the vortex-tree failure location equation is a circle whose roots are solved from a quadratic, as described by Letzmann (1923). However, as φ decreases towards 0.5, the shape of the vortex-tree failure location equation deforms from a circle to an elliptical-like shape, which for an arbitrary φ cannot be expressed explicitly with y as a function of x. Moreover, for an arbitrary value of φ, the specific form cannot always be accurately approximated using the rational polynomials described above, without leading to equation (19) being a higher degree polynomial (20+), which can create significant floating-point errors and be considerably slower (Yuksel 2022). Alternatively, the Rankine profile can be solved in polar form, where r is a function θ (Chen and Lombardo, 2019), followed by some form of interpolation to approximate y for a given x value. However, for the purposes of this study it should be noted that although generally slower, equation (19) can be solved for the Rankine profile (and any other profile functions) using numerical root-finding algorithms. For example, Brent's root finding method (Brent 1972) can be reliably used provided that the roots are bracketed based on their locations relative to the R_{max} circle, as further discussed below.

c. Solving for a Simulated Treefall Pattern

Once the vortex-tree failure location function has been determined, it can be discretely sampled at evenly spaced intervals, as shown in Figure 6, consistent with the spacing of the observed treefall pattern, (Δx) discussed in section 4b. Then, the pattern can be solved by computing the unit vectors at the corresponding locations in the vortex field. However, as discussed in Rhee and Lombardo (2018), to ensure alignment between the observed and simulated patterns the vortex-tree failure location function samples must be offset from the axis of convergence of the pattern’s vectors, and only on the interval where the vortex-tree failure location function has real solutions. The axis of convergence, x_c, can be precomputed by solving for the x-value where the vortex-tree failure location function produces vectors whose x-coordinate transitions between positive and negative. This can be modelled as a numerical root-finding problem that can be solved efficiently using algorithms such as the previously discussed Brent’s method (Brent 1972). Similarly, the length of the pattern can be precomputed by determining the interval, $[a, b]$, over which real solutions exist to the vortex-tree failure location function. This can be solved using a binary (bisection) search over the two intervals, $(-\infty, x^*), (x^*, \infty)$, as any solutions outside the interval, $[a, b]$, will be complex. Equation (23) demonstrates the sampling values of x over the required interval. The values of L_a, L_b represent the lengths of the treefall pattern on either side of the axis of convergence.

$$x = x_c + k\Delta x, \qquad a \le x \le b, \qquad k \in \mathbb{Z} \tag{23}$$

$$L_a = x_c - a, \quad L_b = b - x_c \tag{24}$$

Additionally, by translating a vortex with constant parameters, the trees will continue to fail in the same directions, repeating the same pattern, as shown by Holland et al (2006) and Karstens et al. (2013). As a result, the translated treefall pattern vectors can be projected into a single one-dimensional treefall pattern as shown in Figure 6, similar to an observed treefall pattern created from a transect as shown in Rhee and Lombardo (2018).

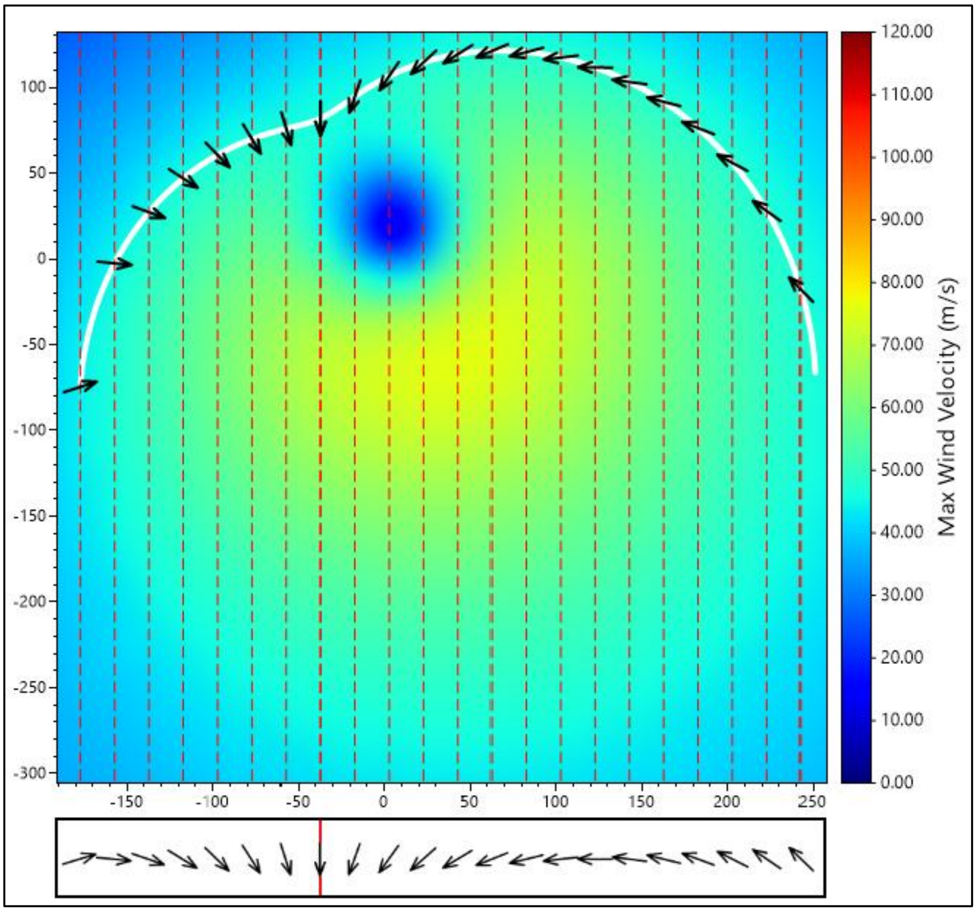

Figure 6: A treefall pattern produced by discretely sampling the vortex-tree failure location function shown in Figure 5 (with $R_{max} = 80m$ instead of $150m$). The red line on the pattern indicates the axis of convergence, x_c.

d. Non-dimensionality of the Vortex-Tree Failure Location Function

As can be seen by examining equation (18) for different unitless profiles, equations (7-11), changing R_{max} causes a proportional stretch to the vortex size and resulting pattern length, but does not affect the underlying shape of the profile or vortex-tree failure location function. Therefore, if the same number of vectors in the pattern are maintained for different R_{max} values, the directions of the pattern's vectors will remain the same, but the overall pattern length will vary. As a result, the same pattern in terms of the number of vectors and their directions can be constructed at any length by changing the R_{max} value.

Additionally, by examining equation (18), a simultaneous proportional change to the velocity components, $V_r, V_t, V_s, V_c \rightarrow \lambda V_r, \lambda V_t, \lambda V_s, \lambda V_c$, would cancel out, having no effect on the shape of the vortex-tree failure location function. Such a change would affect the pattern vector magnitudes but not the directions and, since the treefall pattern is made up of unit vectors, does not affect the produced pattern. Overall, this indicates that a simulated pattern's directions depend only on the various ratios of $V_r{:}V_t{:}V_s{:}V_c$. To accurately estimate a dimensional value of V_{max} (which, from equation (13), depends on V_r, V_t, and V_s), an estimate for V_c and V_s are required. Specific examples and further illustrations of the importance of these ratios can be found throughout the literature. Most notably, α, mentioned previously, and $G_{max} = \sqrt{V_r^2 V_s^{-2} + V_t^2 V_s^{-2}}$, shown by Letzmann (1923) and discussed in Holland et al. (2006), as well as $\eta = V_c V_s^{-1}$, discussed in Chen and Lombardo (2019).

e. Discussion of Analytical Vortex-Tree Failure Patterns

In this section, we examine the typical solutions of equations (18-19) and the implied treefall patterns. By examining the vortex-tree failure locations relative to the $r = R_{max}$ circle, we can classify solutions to the vortex-tree failure location function, equation (18), as "Inner" or "Outer" depending on whether they occur inside or outside the R_{max} circle, respectively. Moreover, patterns can be distinguished by discontinuous transitions between Inner and Outer solutions, described as "Inner-Outer" and "Outer-Inner-Outer" types based on the transitions from left to right in the pattern. Figure 7 shows the four main types of patterns separable by these classifications, which align similarly to the types of patterns first described by Letzmann (1923). Additionally, for single-core, counterclockwise-rotating (cyclonic in the Northern Hemisphere) vortices, only certain combinations of Inner and Outer solutions are possible with the models utilized in this study. For example, the reverse "Outer-Inner" pattern could only occur if clockwise (anti-cyclonic) rotation was utilized.

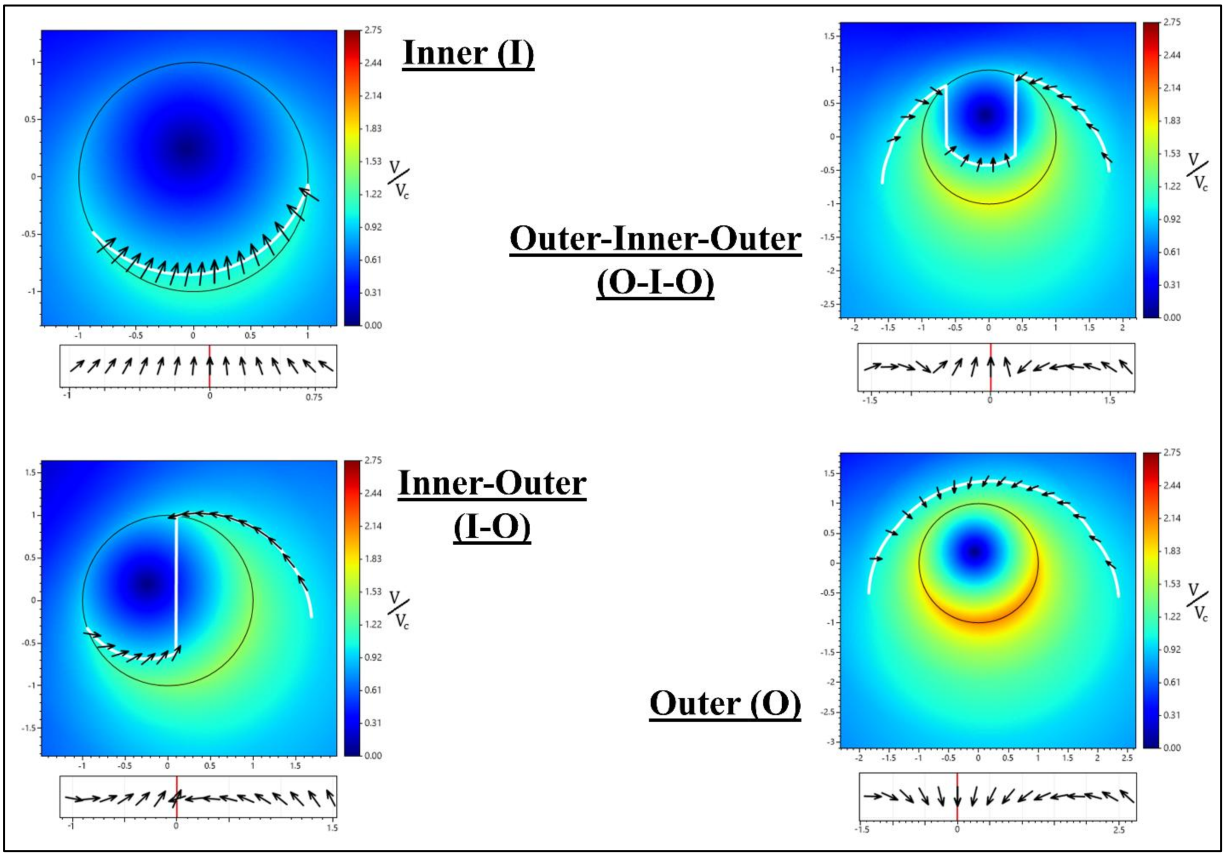

Figure 7: Examples of the four types of patterns which can be classified by vortex-tree failure locations relative to the vortex's R_{max} circle. "Inner" and "Outer" represent solutions inside and outside the R_{max} circle, respectively. The black circle in each graph represents the R_{max} circle. The white curves in each graph represent all vortex-tree failure locations. The red line on each pattern indicates the axis of convergence, x_c.

After analyzing the Inner and Outer partitions for the profiles discussed in this study it appears that generally Inner vectors rotate counterclockwise while Outer vectors rotate clockwise when comparing adjacent vectors from left to right. Moreover, if the axis of convergence happens during an outer partition, the converging treefall vectors will be directed opposite to the translation direction. Additionally, in the case of Inner-Outer patterns, the axis of convergence is taken as the axis where the discontinuity between Inner and Outer solutions occurs and is denoted in Figure 7 with two vectors on either side of the discontinuity. Importantly, their rotation and convergence directions can be utilized to visually distinguish Inner and Outer partitions in observed treefall patterns.

When examining the modelled V_{max} for fixed estimates of V_c and V_s, several observations can be made. For $V_{max} < V_c$, obviously no pattern will be produced since no trees will be downed. As shown in Figure 8, once $V_{max} > V_c$, trees will fail. By considering an increasing value of V_{max}, an Inner (I) type pattern will develop first. As V_{max} is increased further, the pattern transitions from Inner to one of the combinational type patterns, Outer-Inner-Outer (O-I-O), or Inner-Outer (I-O), and finally to the Outer (O) type pattern. Additionally, increasing the swirl ratio S, increases the offset of the pattern's axis of convergence from the center, as shown in Figures 4 and 8. As a result, generally greater swirl ratios tend to produce Inner-Outer over Outer-Inner-Outer patterns, but the specific values that are required vary based on the vortex profiles used. Overall, this suggests that different patterns are a good indicator of V_{max} and swirl ratio values when V_c and V_s are known. Additionally, seeing these transitions between pattern types or shifts of the axis of convergence in observed patterns is an indication of changing vortex wind speeds or critical-tree failure speeds, and swirl ratio, respectively, as described (in part) by Holland et al. (2006) and Rhee and Lombardo (2018).

However, for Outer patterns, further increase to V_{max}, eventually results in a convergence of the treefall pattern to a specific Outer type pattern based on the swirl ratio, with only the pattern's length increasing, as can be seen in Figure 8. This is due to the V_r: V_t: V_s: V_c ratio discussed previously becoming dependent on V_r and V_t as $V_r, V_t \gg V_s, V_c$ for a fixed V_s, and V_c. Furthermore, as previously discussed, decreasing the R_{max} value decreases the pattern's length while maintaining the pattern's vector directions. As such, for Outer patterns, often increasing V_{max}, but decreasing R_{max} results in the same pattern. Therefore, in cases where the observed pattern matches closely with an Outer pattern, no upper bound on V_{max} can be obtained without further restrictions imposed on vortex parameters such as a good estimate for R_{max} to limit the simulated pattern's length. Estimating R_{max} can be accomplished using structural damage (Kuligowski et al. 2014; Lombardo et al. 2015). Nevertheless, when only tree damage is present, this problem of matching Outer patterns often limits a practical estimate of V_{max} for Outer type patterns to the minimum required speed for simulating a similar pattern. This can be problematic for estimating the wind speed of EF4 or EF5 tornadoes, as Outer type patterns require the highest overall wind speeds compared to the other types of patterns for fixed, estimated V_c and V_s values.

Although there is no upper bound on V_{max} for Outer type patterns, a lower bound can be found by examining when the vortex-tree failure locations transition between Inner and Outer types. Under the assumption that the vortex-tree failure location function, equation (19), is continuous for Outer type patterns, this problem can generally be reformulated as determining when the minimum wind speed along the R_{max} circle exceeds V_c. This assumption holds for the profiles considered in this study when excluding unlikely conditions such as high swirl ratios ($S \gg 1$) or as $V_s \rightarrow V_c$, further discussed below. Using a similar logic to the calculation for maximum wind speed, it can be shown that Outer type patterns only occur when,

$$V_{max} \geq V_c + 2V_s \tag{25}$$

Overall, this indicates the importance of having accurate estimates of V_c and V_s, when estimating V_{max} for Outer type patterns.

Another notable issue occurs as $V_s \rightarrow V_c$. In this case, the trees can fail solely based on and in the direction of $\overrightarrow{V_s}$, with little influence from V_r or V_t, potentially causing a pattern with little to no convergence. Furthermore, although unlikely, if V_s exceeds V_c, the model breaks down completely as the vortex-tree failure locations become the entire real xy-plane. A potential solution to this problem could be to use decaying translational speeds farther out from the center of the vortex similar to the radial and tangential profiles, as discussed in Lombardo et al. (2023). This is not considered in the current study.

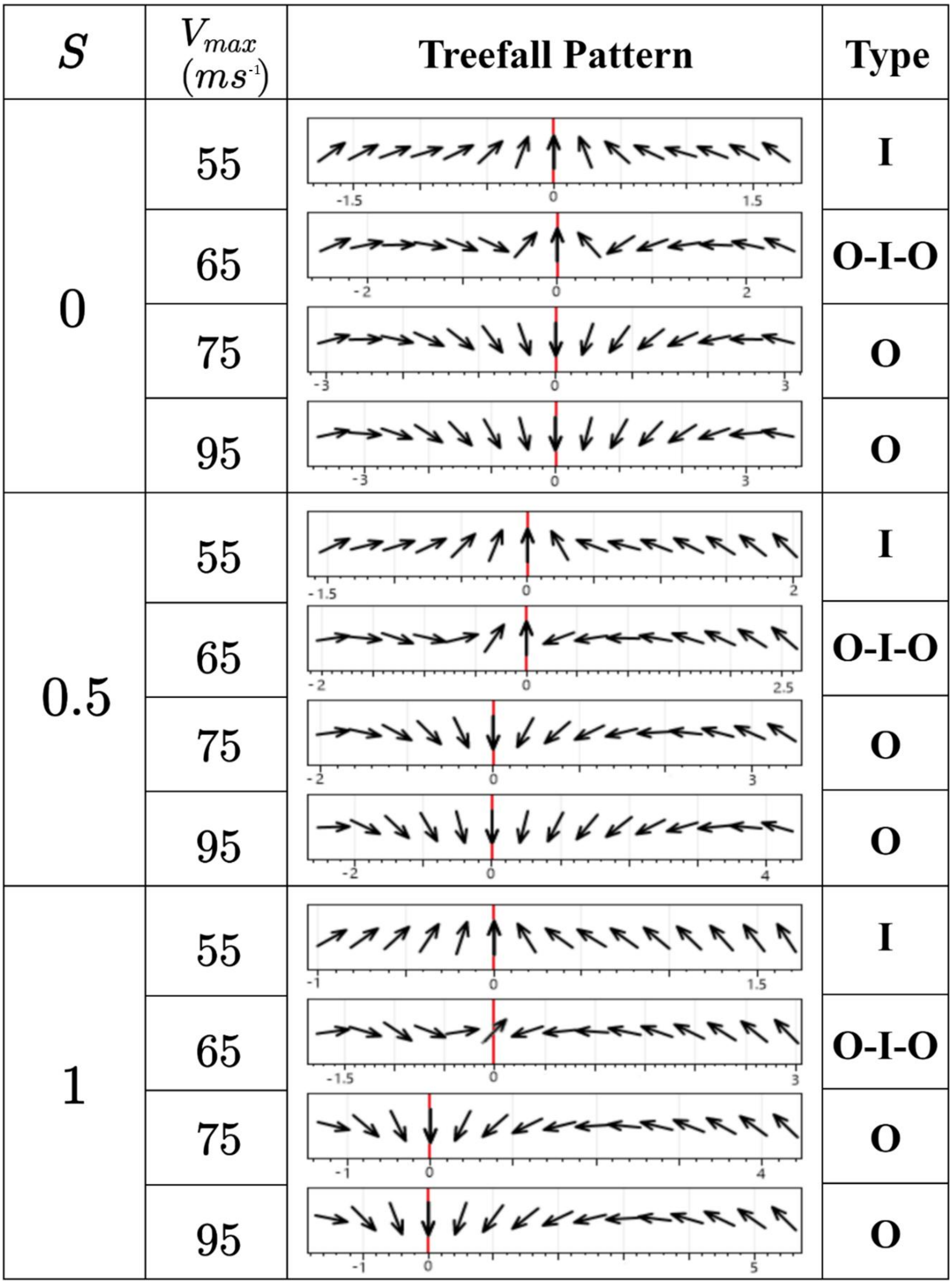

Figure 8: Treefall patterns produced using the Baker-Sterling (Bjerknes) model for different swirl ratios (S) and maximum wind speeds (V_{max}), given a constant $R_{max} = 1m$, $V_c = 40ms^{-1}$, and $V_s = 15ms^{-1}$. For consistency the spacing between vectors (Δx) for a pattern were chosen to create 15 vectors each. The letters "I" and "O" represent "Inner" and "Outer" type vortex-tree failure location solutions respectively.

4. Treefall Pattern Matching

a. Defining Similarity Between Patterns

When comparing observed and analytical patterns, an error metric must be chosen to define how similar the patterns are to each other. In the simplest case, the average of the differences in angles between the vectors can be compared either directly, or using their dot product (Rhee and Lombardo, 2018), with lower averages (error) indicating a better match. In addition to comparing the directions of individual pattern vectors, the overall length and individual lengths on either side of the axis of convergence must be compared to ensure the observed and simulated patterns and corresponding tree damage are consistent in size. This can be accomplished by adding a penalty related to the difference in pattern lengths, as discussed by Rhee and Lombardo (2018). However, for computational efficiency, only comparing vortices with parameters that generate patterns with closely matching lengths can be performed by pre-computing the lengths of patterns. Additionally, there is the question of whether all pattern vectors are equally important. As implied by the four types of patterns in Figure 7, often the vectors closer to the axis of convergence give a better indication of the type of pattern. For example, when comparing Outer and Outer-Inner-Outer patterns, often the only difference occurs with the vectors closer to the axis of convergence. Moreover, the wind speed of tornadoes becomes weaker farther out from the center of the vortex, leading to potentially less reliable average treefall vectors, as fewer trees fail. As such, a linear weighting scheme, $w(x)$, based on distance from the axis of convergence can be utilized, with:

$$w(x) = \begin{cases} 1 - \dfrac{x_c - x}{\Delta x + L_a}, & x \leq x_c \\ 1 - \dfrac{x - x_c}{\Delta x + L_b}, & x > x_c \end{cases} \tag{26}$$

In practice, when comparing the best matching pattern to other close matching patterns, there is often a negligible difference in the pattern error metrics, but some significant differences in vortex characteristics including V_{max}. Often this can be caused by a pattern depending on the various ratios of $V_r : V_t : V_s : V_c$, with different values producing similar ratios. As a result, selecting any of these patterns may yield a different but plausible representation of the observed vortex characteristics. This uncertainty should be considered in tornado damage assessments. To do this the distribution of tornado wind field parameters which produce patterns that match with errors below a particular cut-off threshold can be utilized. This distribution can then be analyzed and taken into account when rating the tornado. However, as discussed previously, using the minimum required wind speed for simulating a similar pattern is effectively required for Outer type patterns when only tree damage is available. Therefore, in the current study, the minimum required wind speed of all plausible patterns must be used.

Determining an ideal cutoff threshold depends heavily on the similarity metric utilized for pattern comparison, as well as human judgment. Figure 9 demonstrates four different pattern matches with increasing pattern error (i.e., decreased similarity) when compared to the same observed pattern using the weighted average of vector angle differences described above, normalized between 0 – 1 by dividing by π. As can be seen in Figure 9, an error of 0.1 or less tends to empirically yield closely matching patterns; however, with errors around 0.2, the simulated pattern no longer produces a reasonable match as the observed and simulated patterns often differ in types (in this case, Outer vs Inner-Outer). Furthermore, if the best-matching simulated patterns do not match with errors below the threshold, then either the observed pattern was poorly chosen, or the observed pattern cannot be reasonably modelled by the analytical single core vortices, indicating that this method may not be suitable for the tornado in question. It is proposed that this could occur when the underlying assumptions do not hold, for example, in the treefall of multi-vortex tornadoes with high swirl ratios, tornadoes with rapidly changing direction or parameters, sections with combined tornadic and downburst damage, or sections with little to no convergence of treefall vectors. Particularly, the case of little to no convergence as the treefall vectors are all falling in the same direction, can occur with weak Inner type patterns, or where V_s approaches the speed of V_c, as described previously. Moreover, by extension, seeing a clear treefall pattern similar to those that can be simulated indicates that the assumptions likely hold, given the earlier, published validations of the overall approach.

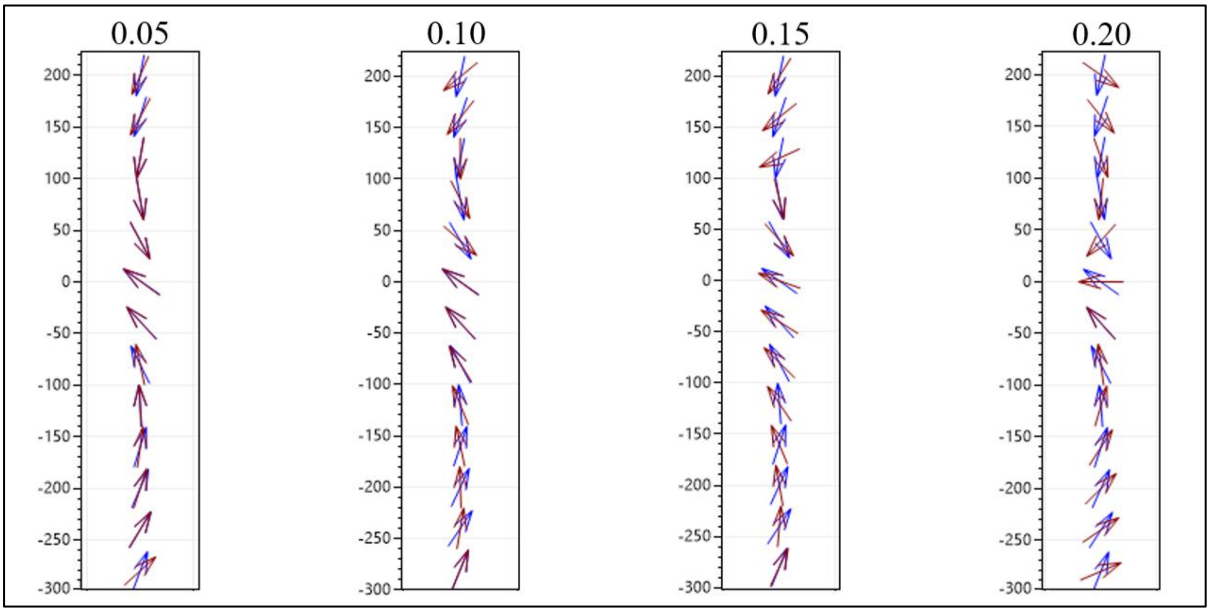

Figure 9: Comparison of 4 different simulated treefall patterns (red) to the same observed pattern (blue), with increasing error (pattern dissimilarity). The errors (0.05 – 0.2) between patterns is calculated using the weighted average of vector angle differences normalized between 0 - 1.

b. Treefall Pattern Matching Software

Using custom treefall pattern matching software designed by the Canadian Severe Storms Laboratory, with a screenshot shown in Figure 10, an observed treefall pattern can be extracted and automatically matched to simulated patterns. The software allows for the placement and adjustment of a transect fit perpendicular to a section of the convergence line. Adjustments include the vector spacing (Δx), transect width, length above and below the convergence line, as well as height and rotation offsets to adjust the transect relative to the underlying section of the convergence line if it is not fit perfectly. As adjustments are made, the treefall vectors inside the transect are shown, along with the corresponding area-averaged treefall pattern given a user-specified treefall pattern vector spacing. A transect is placed and adjusted until the observed treefall pattern is similar to one of the types shown in Figure 7. Moreover, the transect must be fit to the extent of the full length of the observed treefall damage, and with a width ranging generally between 100 – 200 meters. Furthermore, the vector spacing (Δx), should be selected to create a pattern with around 10 – 20 vectors. Generally, fewer than 10 vectors may not fully capture the underlying pattern, and more than 20 are generally unnecessary. Notably, in the software, these patterns are rotated 90 degrees clockwise from those discussed previously (and shown in the earlier figures).

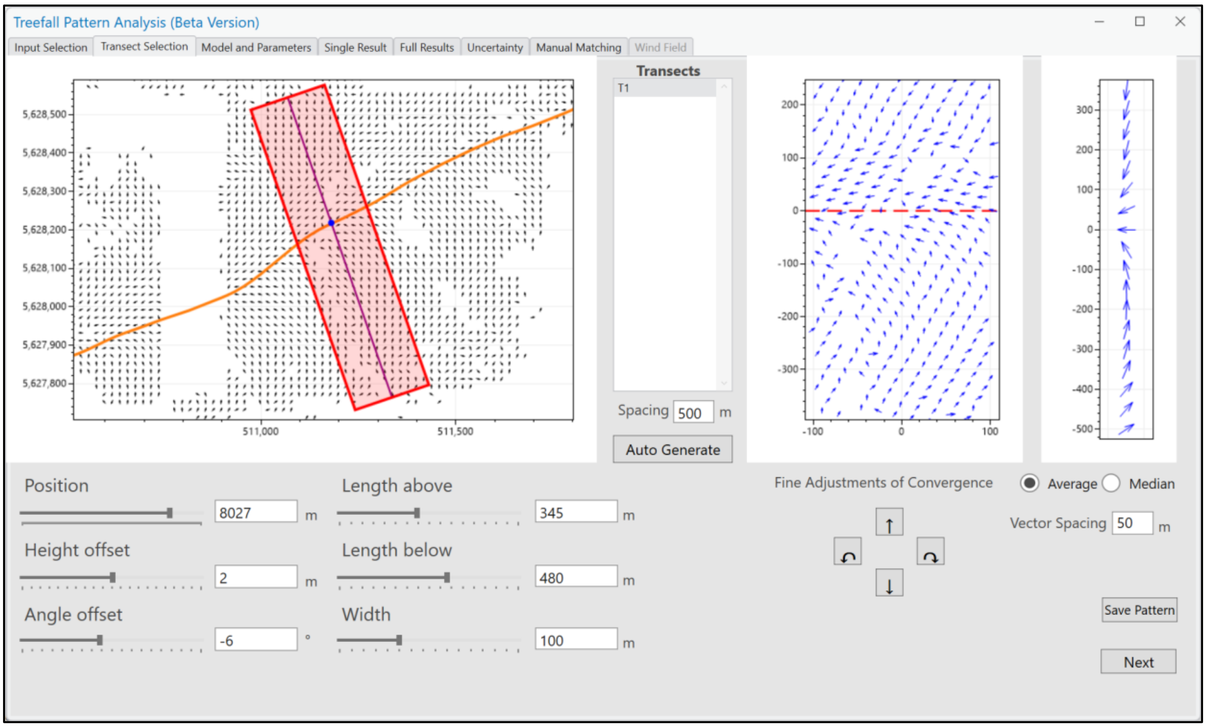

Figure 10: Treefall patterning analysis and matching software, used to find and match treefall patterns. The orange curve is the convergence line, and the red box outlines the selected treefall pattern transect. The graph on the far right shows the area-averaged treefall pattern.

Once a transect is fit, and an observed pattern determined, the user enters the range of parameters $(V_r, V_t, V_s, V_c. R_{max}, \varphi)$ for which corresponding simulated patterns will be generated. For this study, the ranges for V_s and V_c are specified for each event in Tables 6 and 7. Table 3 lists the range and discrete spacing utilized for each other parameter. These ranges were selected to ensure coverage of a wide range of treefall patterns while reducing the required computational time. The range of R_{max} values were selected in common with Miller et al. (2024), based on analysis done by Twisdale et al. (2023). To find the plausible matches, every combination of these parameters is first checked by precomputing the sets of parameters that will produce a pattern with lengths within a one vector spacing of the observed pattern's lengths. Then, for each of these sets of parameters, a simulated pattern is produced, and its error relative to the observed pattern is calculated. Sets of parameters with errors less than the chosen threshold of 0.1 are recorded. Once all these sets of parameters are compared, the overall best matching parameters, as well as the minimum V_{3-max} needed to produce a similar pattern, can be determined.

Table 3: Parameters ranges and discrete sampling spacing used by the treefall pattern matching software.

Parameter	**Range**	**Discrete Spacing**
V_r	20 – 80 ms^{-1}	1 ms^{-1}
V_t	1 – 50 ms^{-1}	1 ms^{-1}
R_{max}	0.1 – 0.5 x Observed pattern length	5 m
φ (Rankine)	0.50 – 1.00	0.05

c. Accounting for Uncertainty

During the process of finding and matching an observed treefall pattern, there are five main sources of uncertainty:

1. The treefall pattern matching error function and weighting scheme
2. The threshold of error for a reasonably well-matching pattern
3. The placement and dimensions of the transect
4. Estimates for critical tree failure and translational speed parameters (V_c, V_s)
5. Differences between vortex profiles (wind field models)

Although different error functions and weighting schemes for comparing treefall patterns have been utilized in other studies, the current study utilizes the average angle difference and linear weighting scheme for the reasons previously mentioned. Along the same lines, although the threshold for a reasonably well-matched pair of patterns could be changed from 0.1, as shown in Figure 9, 0.1 tends to correspond to reasonably well-matched patterns, while airing on the conservative side in comparison to higher values such as 0.2. Notably, using a lower threshold yields a subset of the matches from a higher threshold.

Along a tornado's damage path through a forest, there will be multiple locations where distinct treefall patterns appear. These can be used to develop contours of estimated wind speed or EF-scale rating (Rhee and Lombardo 2018). As for structural damage, estimates for the maximum wind speeds from treefall patterns are based on the location of damage with the highest associated wind speed estimate. (We note that, while structural damage is a point estimate, the current method uses tornado wind field parameters that consider an entire transect of path.) As such, all-distinct treefall patterns must be tested to ensure the highest wind speed estimate is found. Then, for each distinct region where a different treefall pattern is observed, manual changes to the size and orientation of the transect can change the pattern, adding uncertainty to the estimated wind speed. Most notably, as seen in Figure 8, changes to the pattern lengths can result in significant changes to the best-matching swirl ratios.

As well, the ranges of estimated V_c, and V_s values, along with the use of different vortex profiles (models) yield additional uncertainty. In order to assess the uncertainty of sources 3-5, a Monte Carlo style simulation is utilized in this study to compare 10,000 random variations to transect location / dimensions, vortex profiles, and V_c / V_s values for each analyzed tornado. For each of the 10,000 pattern-matching processes previously described, the transect is randomly manipulated, with a fixed V_c / V_s value selected from the ranges in Tables 6/7 and a vortex model selected from those discussed in section 2b. Moreover, fixing V_c and V_s for each run of the Monte Carlo simulation ensures only unique $V_r\!: V_t\!: V_s\!: V_c$ ratios are tested, improving computational efficiency by avoiding comparing duplicate patterns as discussed above. Table 4 lists the parameters used to randomly adjust the transect in the Monte Carlo simulation and their associated ranges. Figure 11 depicts a flow chart of the overall Monte Carlo simulation methodology used to estimate a tornado's parameters and the associated uncertainty.

Table 4: List of randomized transect parameters and ranges for a Monte Carlo style simulation.

Randomized Transect Parameter	**Offset Range**
Vector spacing (Δx)	$\pm 25\%$
Transect Lengths (L_a, L_b)	$\pm 10\%$
Transect width	$\pm 10\%$
Transect height offset (from convergence line)	$\pm \Delta x$
Transect angle offset (from convergence line)	$\pm 5°$

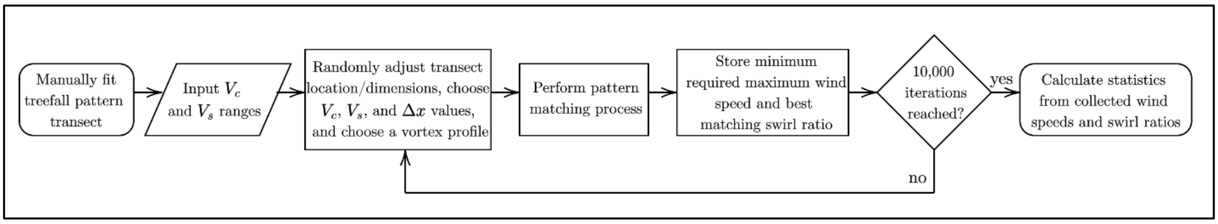

Figure 11: Flow chart describing the Monte Carlo style simulation utilized to measure uncertainty in treefall pattern matching.

5. Application to Observed Tornado Events

a. Overview

For this study, four Canadian forest-based tornadoes were selected for analysis, as listed in Table 5. Events were selected which met the following criteria. Firstly, the events must be primarily forest-based, with a rating of EF2 based on 80% of mature trees snapped or uprooted in the area of worst damage. This is to ensure that enough trees were damaged to warrant the use of treefall patterns. Secondly, there must be visible treefall patterns similar to those of Figures 7-8; otherwise, the assumptions made may not be valid. Thirdly, a preference was given to events with “Outer” type patterns as they tend to produce the highest wind speeds when V_c and V_s have been estimated. For each event, all sections of the damage track with distinct treefall patterns were examined. Only the sections with the highest wind speed estimates are presented here. Additionally, for comparison with the study by Stevenson et al. (2023), the Alonsa, MB tornado, was included.

Table 5: List of tornado events selected for treefall pattern analysis in this study.

Event Name (Location)	**Date**	**Path Length (km)**	**Max Path Width (m)**	**EF Rating (from NTP)**
Alonsa, MB	3 Aug 2018	15.7	1200	EF4
Lake Traverse, ON	15 July 2021	13.3	530	EF2
Dwight, ON	15 July 2021	7.1	400	EF2
Lac Gus, QC	5 Sept 2018	11.4	450	EF2

All of these events were previously analyzed by the Northern Tornadoes project (NTP) using their standard procedures. Large-scale aerial imagery of the tornado’s track and surrounding area using aircraft is collected (Sills et al. 2020). The collected imagery is 5 cm x

5 cm per pixel in resolution or better. Next, Treefall Identification and Directional Analysis (TrIDA) maps are created, showing a vector field of the treefall directions like those automatically produced using the software in Butt et al. (2024). A tornado convergence line (curve) is then fit along the path where the treefall vectors converge. Finally, the tornado is rated using the Scalable Box Method (Sills et al., 2020). Ratings are included in Table 5. Additionally, V_s and V_c must be determined for each event in order to use the treefall methodology. Table 6 provides the estimates related to V_s while Table 7 provides the estimates related to V_c.

To determine V_s, the available radar data and its resolution are examined. A radius of uncertainty is estimated for each selected point used to determine the center of the mesocyclone, with its variance calculated as this radius squared. The radius of uncertainty was estimated at 1500 m and 3000 m for Doppler radial velocity data and reflectivity data respectively. For the Lac Gus tornado, due to the lower quality of available reflectivity data, an additional 1500 m of uncertainty was added. This uncertainty is assumed to be independent between the selected points, and as such the overall standard deviation of uncertainty can be calculated as the square root of the sum of the estimated variance at each point. Notably, this radius of uncertainty also attempts to account for potential deviations of the actual translational speed from the average translational speed over the short duration of a treefall pattern transect. From both radar and treefall convergence, it is clear that the tornadoes in the events considered travelled along quasi-linear paths, without looping. Table 6 gives a summary of the estimated V_s values and uncertainty (standard deviation) for each event.

Table 6: List of estimated V_s values and data sources for each tornado event analyzed.

Event	Radar Station	Data Type	Total Time (s)	Total Distance (m)	Radius of Uncertainty (m)	Estimated V_s (ms^{-1})
Alonsa, MB	XWL Woodlands, MB	Radial Velocity	1200	13600	1500	11 ± 2
Lake Traverse, ON	WBI Britt, ON	Reflectivity	1800	33700	3000	19 ± 4
Dwight, ON	CASKR King City, ON	Radial Velocity	720	10200	1500	14 ± 2
Lac Gus, QC	WMN Ste-Anne de Bellevue, QC	Reflectivity	1200	18700	4500	16 ± 6

To determine V_c, ground surveys were conducted for each event to determine the dominant tree species and collect soil samples. Additionally, values of DBH (diameter at breast height) and h were determined using drone and aerial imagery of each event location, by sampling 50 – 100 trees around the locations of worst damage. Specifically, trees which were unobstructed and clearly visible were selected to ensure accurate measurements. This revealed that balsam fir (*Abies balsamea*) and paper birch (*Betula papyrifera*) were dominant in the damaged areas and the parameters used in the analysis reflect the relative proportions of each species. Values for D were estimated by taking the square root of the average area around each tree using:

$$D = \sqrt{An^{-1}} \tag{27}$$

where A is the area and n is the number of trees in the fitted treefall pattern transect with the highest estimated wind speed for each event.

The values of SW for each tree species were estimated using the tree taper equation of Honer (1967), with the appropriate shape parameters (b_0 = 2.14 and b_1 = 301.63 for BF and b_0 = 2.22 and b_1 = 300.37 for PB) and the wood density values (ρ_{wood} = 761 and 866 kg/m^3 for BF and PB, respectively) were based on average green heartwood and sapwood moisture contents (Ross, 2010). C_{reg} was estimated using data from Ruel et al (2000) for balsam fir growing in freely draining mesic till (189 Nm/kg) and Krišans et al. (2022) for birch growing in drained mineral soils (190 Nm/kg). Well-known analytical expressions for vegetated surfaces (Raupach, 1994) were used for d/h (0.8) and z_o/h (0.09), as a function of forest canopy height and leaf area index (LAI). The LAIs for balsam fir and paper birch were estimated from field and remote sensing studies of boreal mixed-wood forests in Ontario and Newfoundland (Hunt et al. 1999; Pope & Treitz 2013) and were assumed lie between 1.5 and 3.5 m^{2}m^{-2}. Notably, all events were below 500m in elevation.

Table 7 gives a summary of the characteristics and tree statistics for each event, as well as the estimated V_c values and uncertainty using the methodology described in section 3. Notably, these values match closely to the upper and lower bound wind speeds of 73-113 mph (32 - 51 ms^{-1}) provided by WSEC (2006) for the overturning of softwood trees in the U.S. EF scale and represent 86-95% of separately calculated stem break critical wind speeds for each of the sampled trees (based on the relevant equation from Gardiner et al. 2000). Additionally, for the Alonsa, MB event, the V_c value and uncertainty of 47.5 ± 12.5 ms^{-1}, are taken from Stevenson et al. (2023) for comparison, which used a similar critical tree failure estimation methodology.

Table 7: List of estimated V_c values and associated tree parameters for each analyzed tornado event.

Event	**Dominant Species**	**Average Height (m)**	**Average DBH (m)**	**Average D (m)**	**Estimated V_c (ms^{-1})**
Lake Traverse, ON	Balsam Fir	23 ± 3	0.39 ± 0.08	3.7	44 ± 8
Dwight, ON	Paper Birch	19 ± 3	0.31 ± 0.09	3.6	39 ± 10
Lac Gus, QC	Balsam Fir / Paper Birch (50:50)	14 ± 3 / 16 ± 3	0.31 ± 0.09 / 0.40 ± 0.10	3.9	31 ± 8

b. Results

The results for this study are listed in Table 8 and Figures 12-15. For each event, the estimated minimum required V_{3-max} as well as the best matching swirl ratio (S) are given as the median result with one standard deviation of uncertainty. These values are based on 10,000 runs of each associated Monte Carlo simulation. The median was chosen over the mean as it is more resistant to outliers and due to the distributions being slightly skewed from symmetric normal distributions, as can be seen in Figures 12 – 15 for the four analyzed events.

Figure 16 shows the selected treefall patterns for each event compared to the best-matching simulated pattern. As can be seen, all selected treefall patterns were of the Outer type, and are matched closely by the simulated patterns, below the 0.1 weighted average threshold described previously. For the Alonsa event, the median V_{3-max} of 75 ms^{-1} matches the current rating of EF4 based on structural damage; however, is approximately 10 ms^{-1} lower than the treefall pattern estimates from Stevenson et al. (2023), indicative of the more conservative nature of this study caused by the utilization of the minimum required V_{3-max}, instead of directly estimating the instantaneous maximum wind speed. Nevertheless, two of the other three newly analyzed tornadoes produced estimated ratings above their current rating of EF2. Furthermore, in the case of the Lake Traverse tornado, every run of the Monte Carlo simulation required a minimum V_{3-max} above the threshold for EF3. This suggests a possible explanation is that many current Canadian EF2 tornadoes in forested areas are underrated due to insufficient higher-end damage indicators, but further research using this methodology will be required to assess this conclusively.

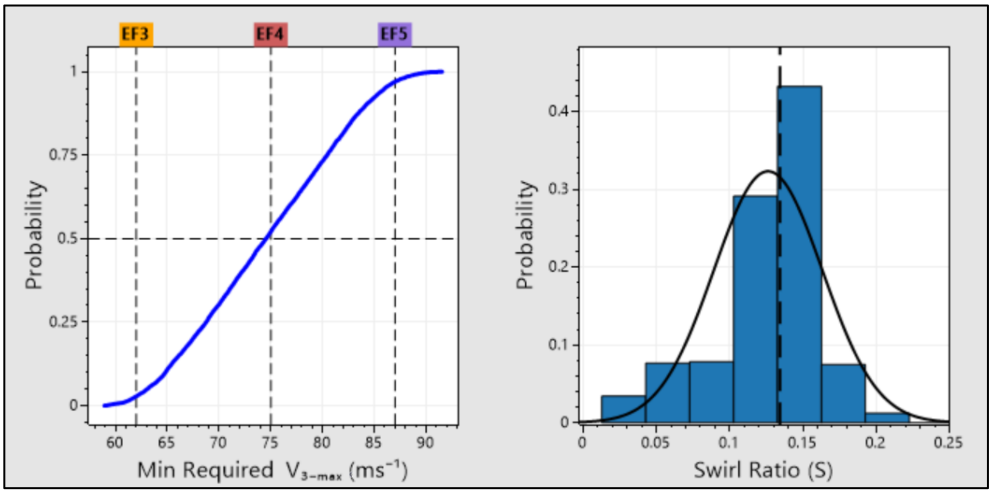

Figure 12: Results from 10,000 runs of the treefall pattern matching Monte Carlo simulation for the Alonsa, MB tornado. For the swirl ratio histogram, the dashed line represents the median result, and the solid line is a fit normal distribution.

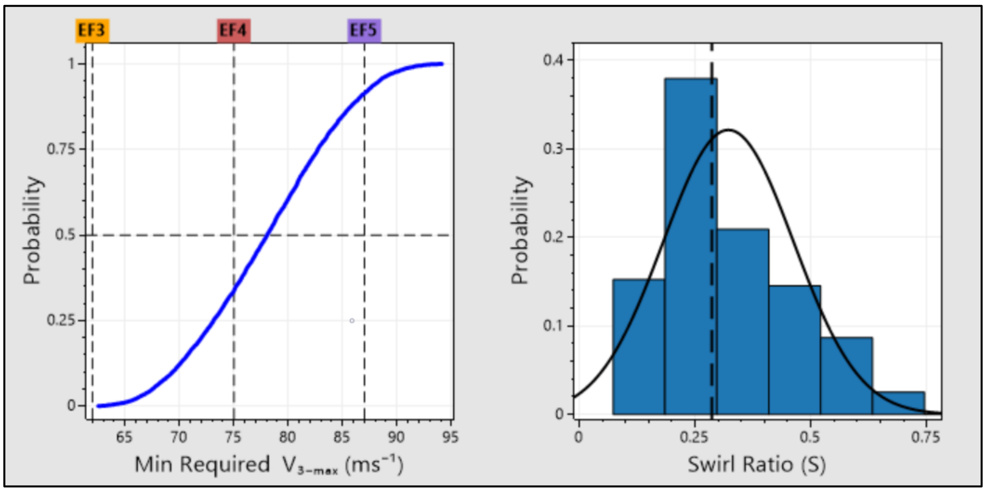

Figure 13: Results from 10,000 runs of the treefall pattern matching Monte Carlo simulation for the Lake Traverse, ON tornado. For the swirl ratio histogram, the dashed line represents the median result, and the solid line is a fit normal distribution.

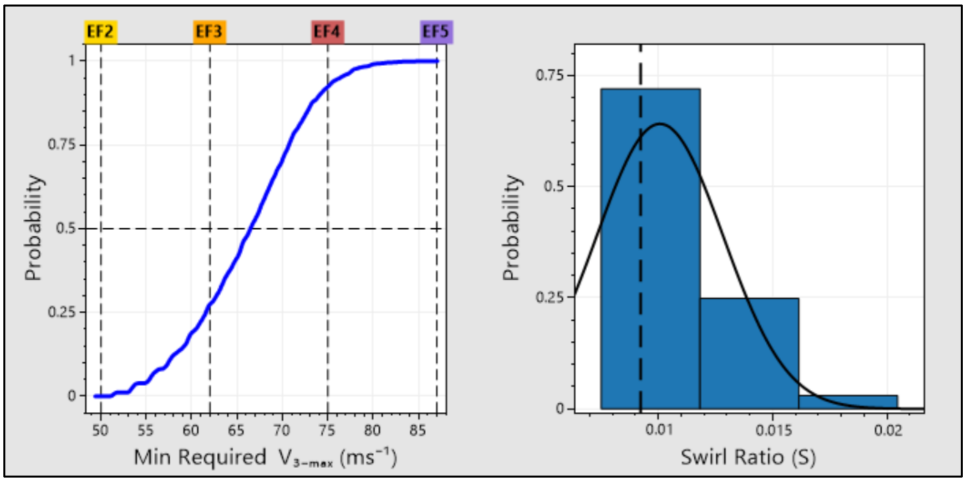

Figure 14: Results from 10,000 runs of the treefall pattern matching Monte Carlo simulation for the Dwight, ON tornado. For the swirl ratio histogram, the dashed line represents the median result, and the solid line is a fit normal distribution.

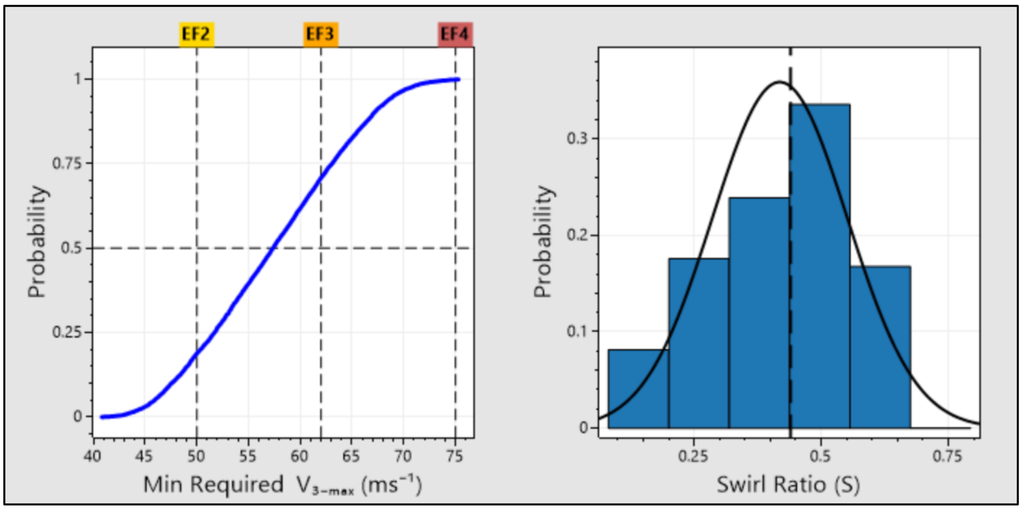

Figure 15: Results from 10,000 runs of the treefall pattern matching Monte Carlo simulation for the Lac Gus, QC tornado. For the swirl ratio histogram, the dashed line represents the median result, and the solid line is a fit normal distribution.

Table 8: Results for the tornadoes analyzed. Results are listed as median ± one standard deviation.

<table>
<tr><th>Event</th><th>V_c (ms^{-1})</th><th>V_s (ms^{-1})</th><th>S</th><th>V_{3-max} (ms^{-1})</th><th>Treefall Pattern Rating</th><th>Canadian EF-Scale Rating</th></tr>
<tr><td>Alonsa, MB, 2018</td><td>48 ± 12</td><td>11 ± 2</td><td>0.13 ± 0.04</td><td>75 ± 9</td><td>EF4</td><td>EF4</td></tr>
<tr><td>Lake Traverse, ON, 2021</td><td>44 ± 9</td><td>19 ± 4</td><td>0.29 ± 0.1</td><td>78 ± 7</td><td>EF4</td><td rowspan="3">EF2</td></tr>
<tr><td>Dwight, ON, 2021</td><td>39 ± 10</td><td>14 ± 2</td><td>0.01 ± 0.01</td><td>67 ± 7</td><td>EF3</td></tr>
<tr><td>Lac Gus, QC, 2018</td><td>31 ± 8</td><td>16 ± 6</td><td>0.44 ± 0.1</td><td>57 ± 8</td><td>EF2</td></tr>
</table>

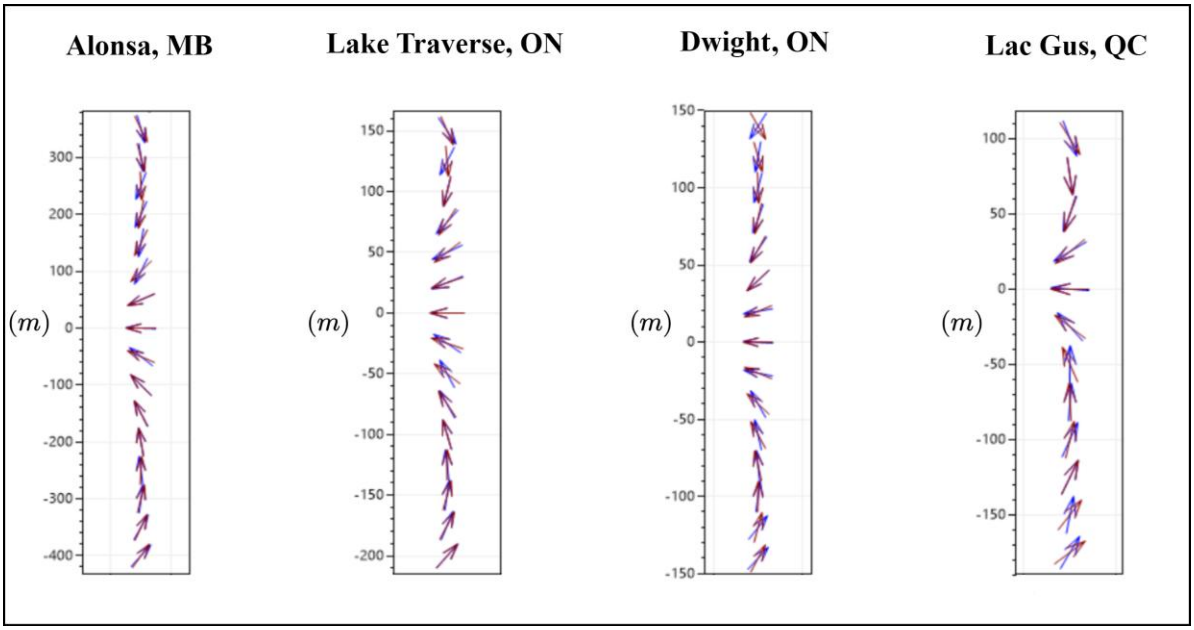

Figure 16: From left to right, observed (blue) vs. simulated (red) treefall patterns for the Alonsa, MB, Lake Traverse, ON, Dwight, ON, and Lac Gus, QC events.

6. Conclusions

This study developed and assessed analytical solutions of tornado treefall patterns and their uncertainty for estimating maximum 3-second gust speeds and swirl ratios. Through mathematical analysis, treefall patterns were determined to depend primarily on the ratio of the radial, tangential, translational, and critical tree failure speeds. As a result, when an estimate for the critical tree failure and tornado translation speeds are known, treefall patterns can be consistently linked to an estimate of a tornado's maximum wind speed and swirl ratio. Four typical types of tornado treefall patterns are differentiated based on vortex-tree failure locations relative to the radius of maximum wind speed. It was found that Outer type patterns, which have convergence in the opposite direction of the translation direction, correlate to the highest wind speeds compared to the other types of patterns for known critical tree failure and translational speed values. However, potential limitations exist for high wind speeds (EF5) due to the discussed Outer pattern convergence, fast travelling tornadoes due to the translational speed approaching or exceeding the critical tree failure speed, and multi-vortex tornadoes whose profiles differ from those discussed in this study.

Four forest-based tornado events, including the previously analyzed Alonsa EF4 tornado, were selected with assessments of their critical tree failure and translation speeds values performed. By utilizing the Monte Carlo simulation methodology, two of the three newly analyzed events were estimated to have higher EF-scale ratings than those suggested by the Scalable Box Method currently used in the Canadian EF scale. Future work will apply this methodology to all Canadian forest tornadoes with significant treefall damage, since it has the potential to change the ratings for many higher intensity tornadoes.

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Data Availability Statement.

All the open-access data used in this study can be obtained from the Northern Tornadoes Project website (www.uwo.ca/ntp). An ArcGIS Pro add-on along with the code used in the study can be found in our GitHub repository (https://github.com/Canadian-Severe-Storms-Laboratory/Treefall_Pattern_Analysis).

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